Firm–technology knowledge graph to leakage-aware entry prediction (features at year t; label from t+1)

1 Time-stamped patents

USPTO PatentsView grants

Organization assignees only

Grant year 2010–2023

CPC subclass (four characters)

Input: firm_id, year, cpc_code

2 Graph G_t = (F, C, E_t)

Firm–class edges observed ≤ t

Class co-occurrence in portfolios

Candidate set from 2010–2012

No future edges in features

C is fixed; E_t is time-indexed

3 Inspectable features

Portfolio size; activity count

Class prevalence; recent growth

Common-neighbour co-occurrence

Peer adoption; class degree

Graph block added only in M3

4 Same-candidate models

M1 logistic regression

M2 random forest (tabular)

M3 forest + graph features

Thresholds from year 2019

Primary contrast: M3 versus M2

5 Out-of-time evaluation (pre-specified split)

Train 2013–2019

1,540,698 pairs

Fit preprocessing here only

Embargo 2020

No train/test overlap

Label gap year

Test 2021–2022

455,458 pairs; 10,411 firms

Positive rate 1.11%

Primary metric

Average precision

AUC, balanced acc., Brier

Uncertainty

Firm-cluster bootstrap

M3 − M2 ΔAP interval

Design rule: every feature for prediction year t uses records dated ≤ t. Graph complexity is retained only if it improves held-out average precision.